

FRP Backing Rings



FRP backing rings for corrosion-resistant flange assemblies. Manufactured to FRP Depot standard flange compatibility dimensions; standard construction is premium unsaturated polyester resin.

1 – 14 in.

standard catalogue sizes

ANSI 150#

drilling pattern only

Polyester

standard resin system

10 sizes

standard stock range



Top view



Low side angle



Opposite face



Laminate and hole detail



Edge profile

STANDARD NOMINAL SIZES

1" · 1-1/2" · 2" · 3" · 4" · 6" · 8" · 10" · 12" · 14"

STANDARD SPECIFICATION

ANSI B16.5 150# Drilling · Polyester

SELECTION NOTES

- Use the dimensional table on the following page for standard geometry.
- Standard resin is Unsaturated Polyester (PE). Premium vinyl ester or custom resins available upon request.
- Confirm solid or split construction, service rating, and current availability before ordering.

150# describes drilling pattern only.

Dimensions follow ANSI B16.5 Class 150 bolt circle, hole count and hole diameter. It is not an assembled-system pressure rating.

Images are representative marketing views of the product family, not dimensional evidence. FRP Depot standard dimensional specifications and selected-item confirmation control dimensional fit.

Backing Ring Dimensions

Dimensional standard: FRP Depot standard backing ring geometry for loose flanges, covering nominal pipe sizes 1" to 14" with ANSI B16.5 Class 150 drilling.

Size in.	DN mm	PN*	Bore B3 mm	FRP T B5 mm	OD B7 mm	ANSI BCD K mm	Hole d mm	Holes N
1"	25	16	51	19	115	79.2	16	4
1-1/2"	40	16	69	19	150	98.6	16	4
2"	50	16	83	19	165	120.6	19	4
3"	80	10	112	22	200	152.4	19	4
4"	100	10	134	24	229	190.5	19	8
6"	150	10	189	30	285	241.3	22	8
8"	200	10	238	32	343	298.4	22	8
10"	250	10	294	34	406	361.9	25	12
12"	300	10	344	36	483	431.8	25	12
14"	350	6	388	38	533	476.2	29	12

B3
ring bore (inside diameter)

B5
FRP ring thickness

B7
ring outside diameter

K / d / N
bolt circle / hole diameter / hole count

SPECIFICATION NOTES AND DESIGN LIMITS

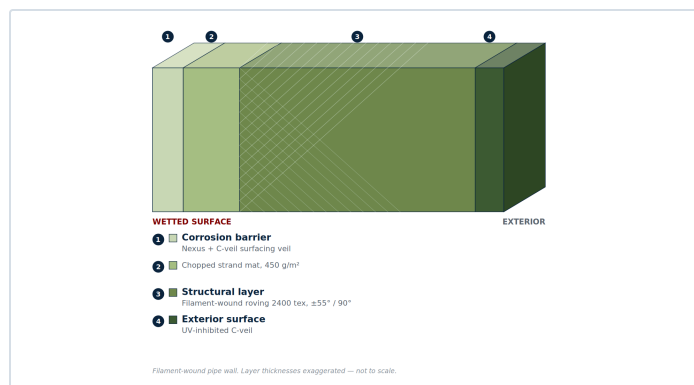
- 1–24 in.: ANSI B16.5 150# drill pattern only. The 150# notation is not a product pressure rating.
- *PN indicates nominal pressure class rating: PN16 at 1–2 in., PN10 at 3–12 in., and PN6 at 14 in.
- Standard construction is solid FRP backing ring with 45° chamfered bore for stub flange radius clearance.
- Standard resin is Unsaturated Polyester (PE). Vinyl ester resins available on request.
- All dimensions are in millimetres (mm) unless specified in inches. Confirm selected-item drawings when fit is critical.

Technical specification for review and selection. Confirm critical dimensions before fabrication.

Resins, Laminate & Standards

Every component is built as a layered laminate. The layer names below are those used by ASTM C582 and ASME RTP-1 and are the common vocabulary of the industry. What differs between products is not the structure but the ply build-up and the way the structural layer is laid down. Wall thicknesses quoted throughout this catalogue are total thicknesses and include the corrosion barrier.

LAMINATE CONSTRUCTION



Filament-wound pipe wall. Layer thicknesses not to scale.

Inner surface

A resin-rich layer reinforced with a surfacing veil — glass, synthetic or carbon. Nominally 0.010–0.020 in. This is the surface the process fluid touches.

Interior layer

Chopped strand mat, two plies minimum. With the inner surface it forms the **corrosion-resistant barrier** — the part of the wall that does the chemical work, and the part excluded from the strength calculation.

Structural layer

Carries the pressure and the mechanical load. Filament wound on pipe; contact moulded — hand lay-up — on every fitting. The first ply against the barrier is always mat.

Outer surface

A resin coat, paraffinated for a full surface cure. UV-inhibited exterior standard on pipe, available on fittings.

Ply-by-ply build-up differs by product and method. The filament-wound pipe schedule is tabulated on page 07; fitting schedules are held by the manufacturer — see the scope note below.

STANDARDS OF MANUFACTURE AND REFERENCE

CGSB 41-GP-22	Process equipment — reinforced polyester, glass fibre. Governing Canadian fabrication standard; covers tanks, pipe and ductwork, and establishes the 10:1 safety factor.
ASTM D5421	Contact-Molded "Fiberglass" Flanges. Governs the stub flanges and manway flanges.
ASME RTP-1	Referenced for bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns only — see the scope note below.
ASME B16.1 Cl. 125	Bolt-hole drilling, NPS 30 and 36.
ASME B18.22.1	Type A Narrow plain washers — governs back-face (BF) diameters.

RESIN SYSTEMS

Derakane 411-350

Bisphenol-A epoxy vinyl ester

HDT	105 °C / 220 °F	TENSILE	86 MPa / 12,000 psi
MODULUS	3,200 MPa	ELONGATION	5–6 %

The standard offering. Broad resistance to acids, alkalis, bleaches and solvents.

Derakane 470-300

Epoxy novolac vinyl ester

HDT	150 °C / 300 °F	TENSILE	85 MPa / 12,500 psi
MODULUS	3,600 MPa	ELONGATION	3–4 %

Specified where elevated temperature, solvent or oxidiser service (chlorine, flue gas) demands higher thermal and chemical resistance.

HDT is a material property, not a service rating. Maximum service temperature is governed by the chemical environment — confirm against the resin manufacturer's chemical resistance guide before specifying.

DESIGN BASIS

- Minimum **10:1 safety factor** on the structural laminate, per CGSB 41-GP-22.
- The corrosion-resistant barrier is excluded from the strength calculation — the safety factor applies to the structural wall only.
- Filament-wound structural layers are designed on a **2:1 hoop-to-axial** stress ratio.

GASKETS & BOLTING

- 1/8 in. full-face elastomeric gasket, Shore A60 ± 5, minimum design seating stress (y) 100 psi, gasket factor (m) 1.
- Flat-face mating flanges only — never bolt FRP to a raised-face metallic flange without a full-face adaptor.
- Tighten in a star pattern in equal stages. FRP flanges are damaged by over-torquing — ask us for torque values.

STANDARDS SCOPE

- ASME RTP-1 is a vessel standard** rated to 15 psig. It does not cover piping or pipe fittings. Where we cite it, we mean bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns — nothing here is offered as an RTP-1 certified vessel component.
- We are a stockist. Item-specific laminate schedules and conformance documentation come from the manufacturer against a specific order — ask at enquiry stage.



FRP DEPOTS

CANADA'S DIRECT SOURCE FOR FRP COMPONENTS

How to order

Send your enquiry to sales@frpdepots.com or call **613-499-4950**. Include the following and we will quote from stock where we can.

- | | |
|---|---|
| 01 Product type and nominal diameter | 02 Pressure rating required |
| 03 Resin system — D411 or D470 | 04 Quantity and length (pipe, couplings) |
| 05 Any non-standard drilling, stub length or joint preparation | 06 Drawings or job specification, if you have them |

Full dimensional data, tolerances and design basis are published in **FRPD-SPEC-001 Revision R3**. Request the current revision with your enquiry.

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